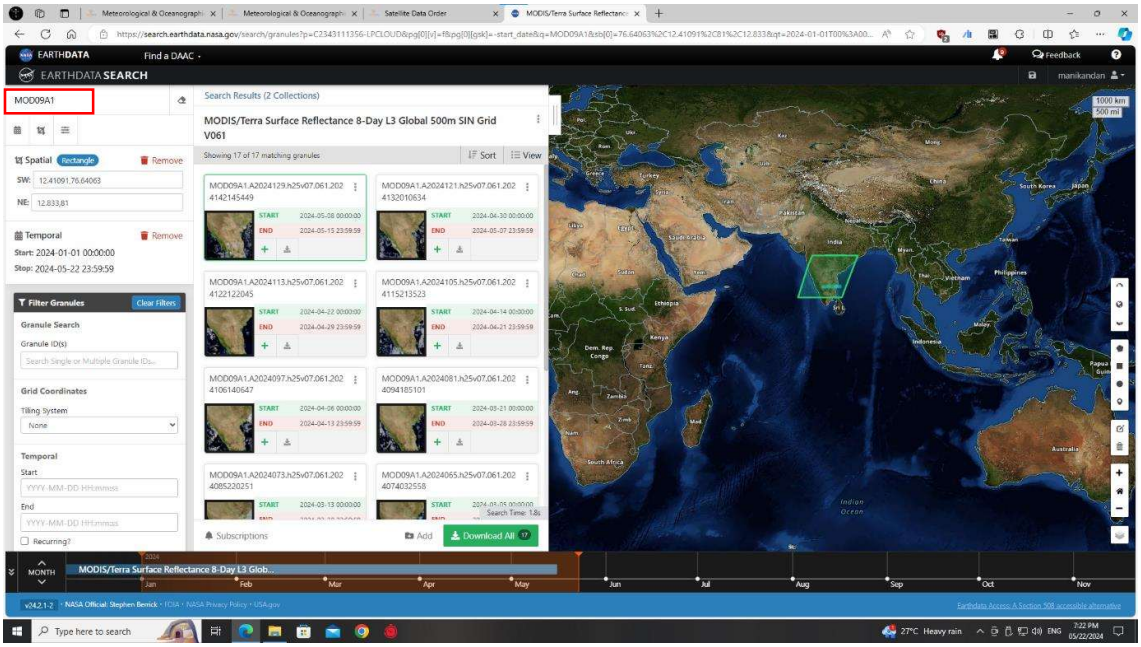
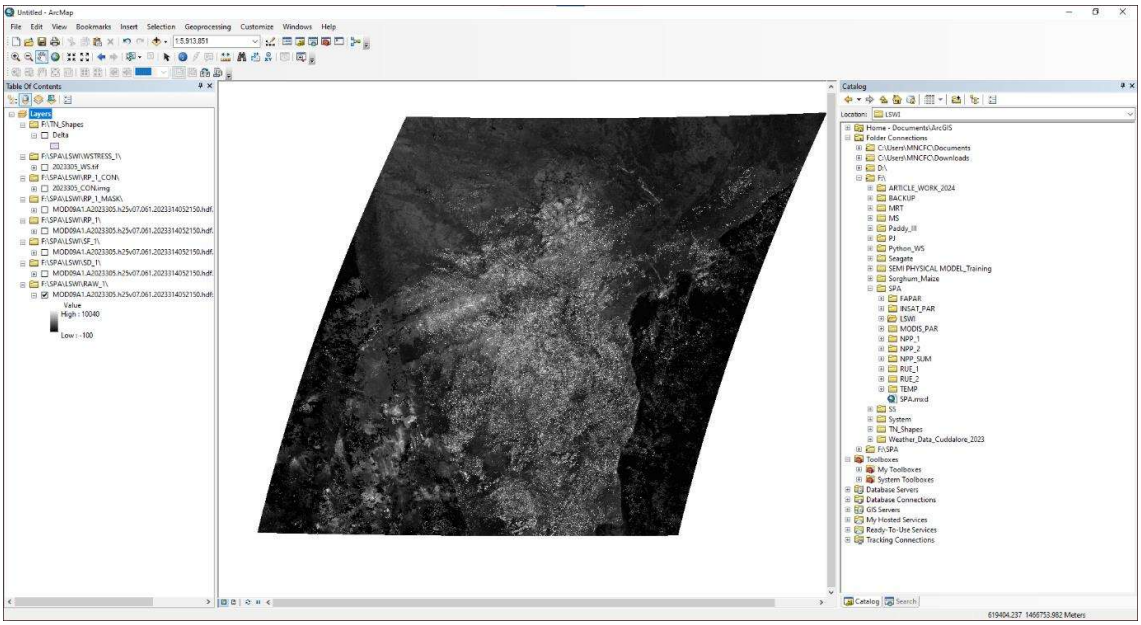
C. Steps for generating Water Stress from LSWI data for study area

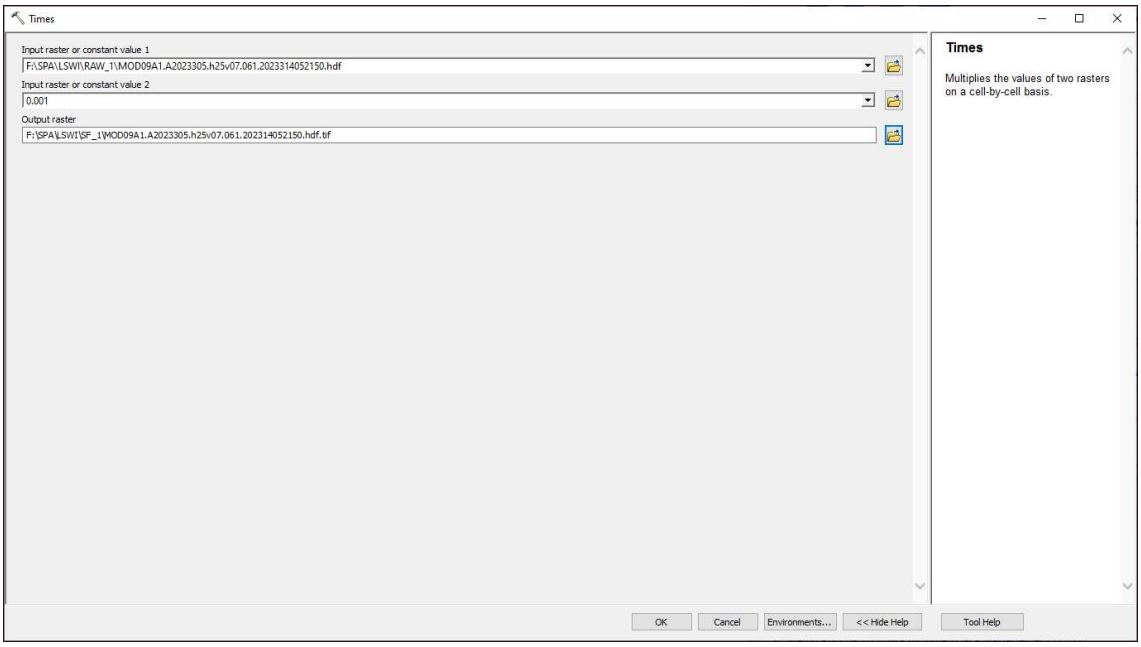
STEP I: MOD09A1 Data acquisition from Earth data search portal



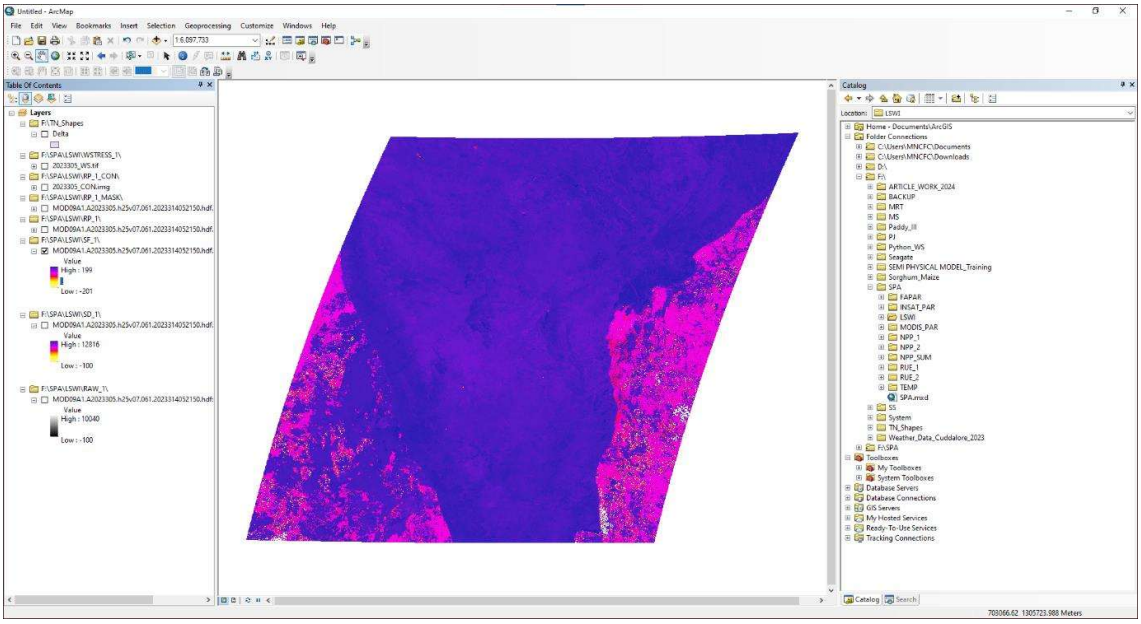
Raw data of MODIS – MOD09A1



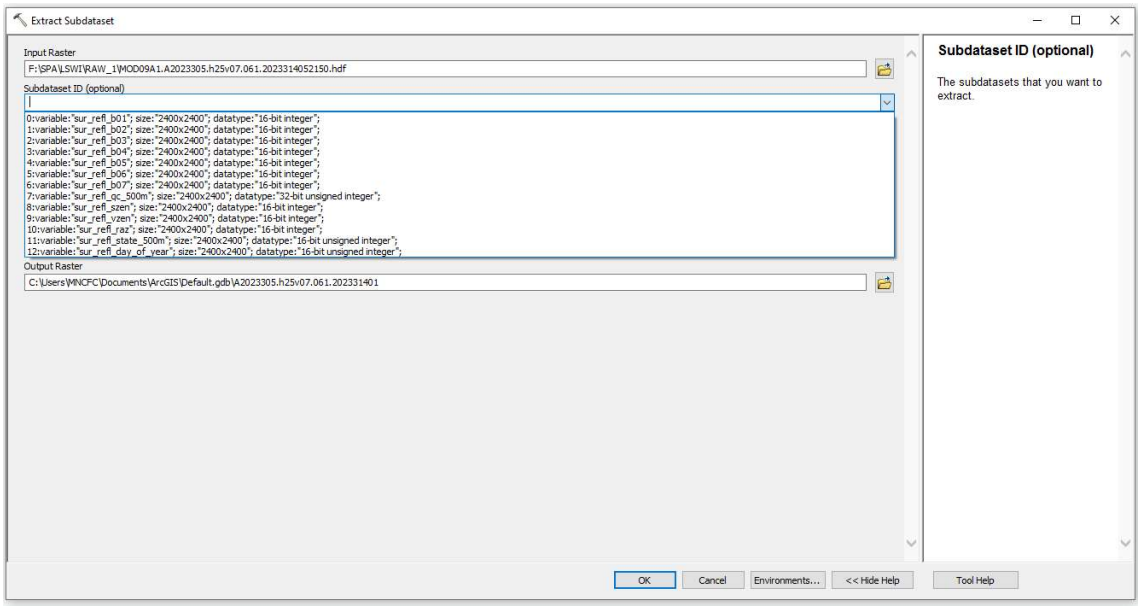
STEP II: Multiplying raster with scale factor (0.0001) for MOD09A1 product using *Times* tool



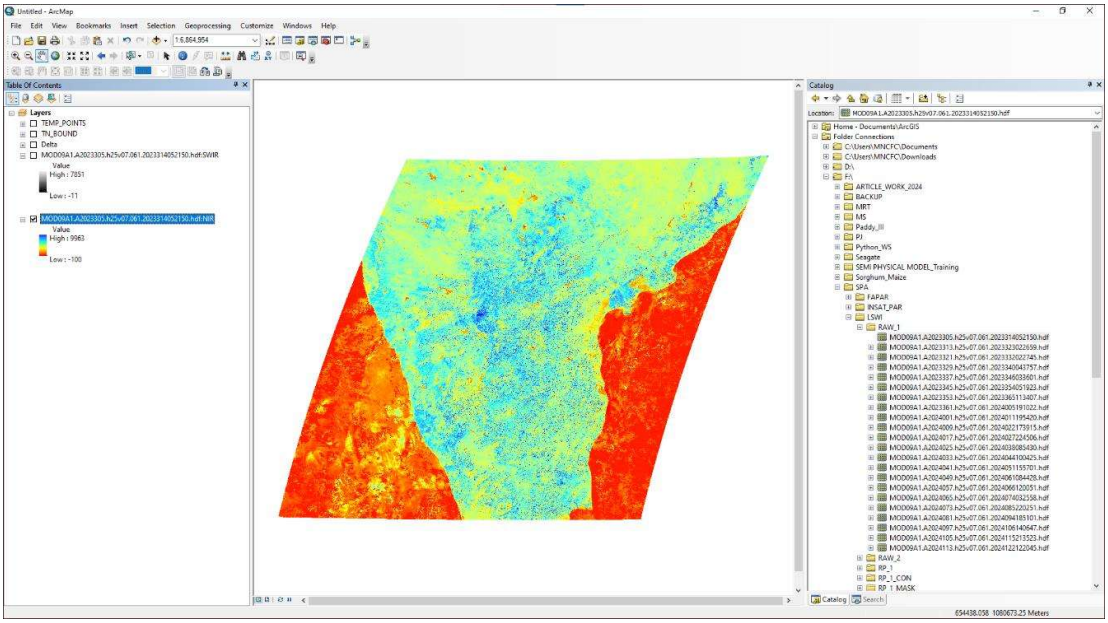
Raster Output of MOD09A1 multiplied with Scale factor



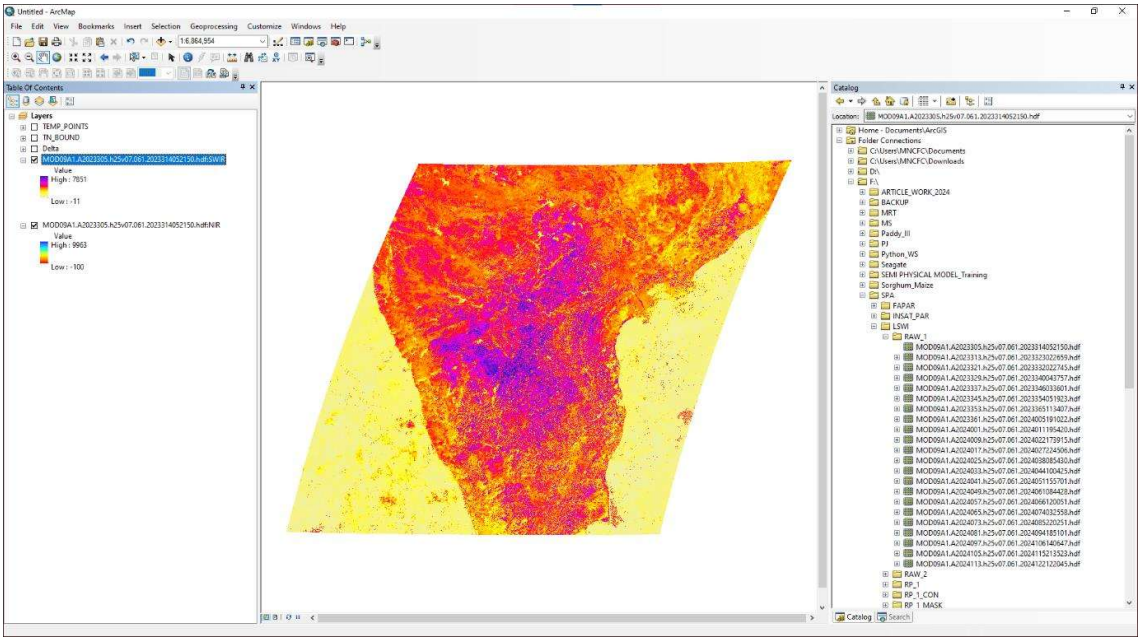
STEP III: Extracting NIR (Band 2) & SWIR (Band 6) from MOD09A1 using *Extract Subdataset* tool



Raster Output of NIR Band from MOD09A1

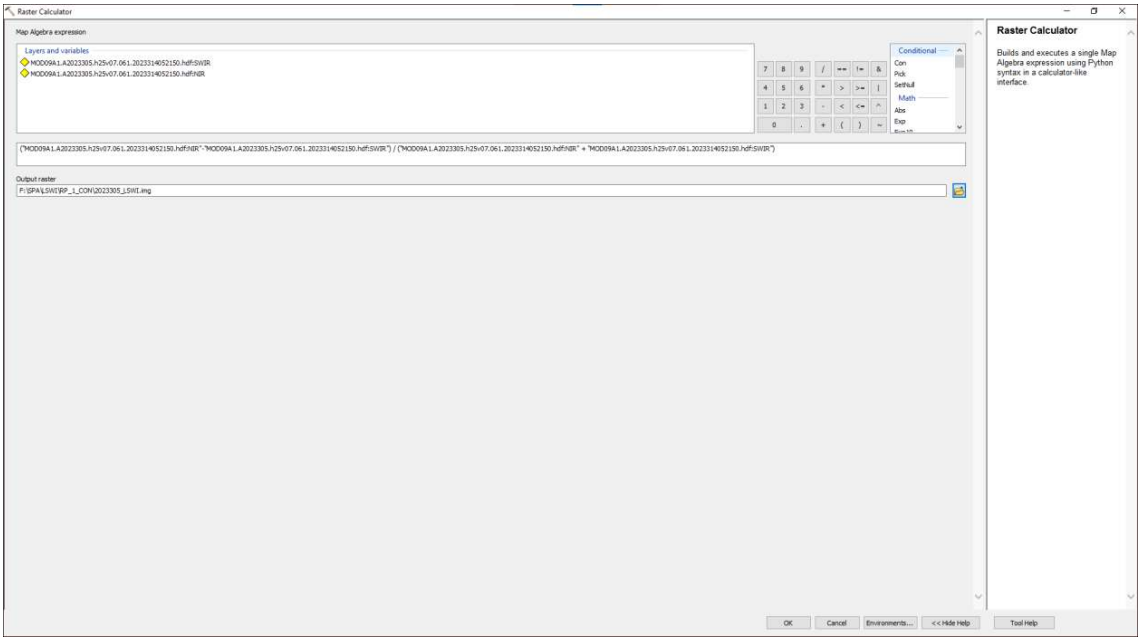


Raster Output of SWIR Band from MOD09A1

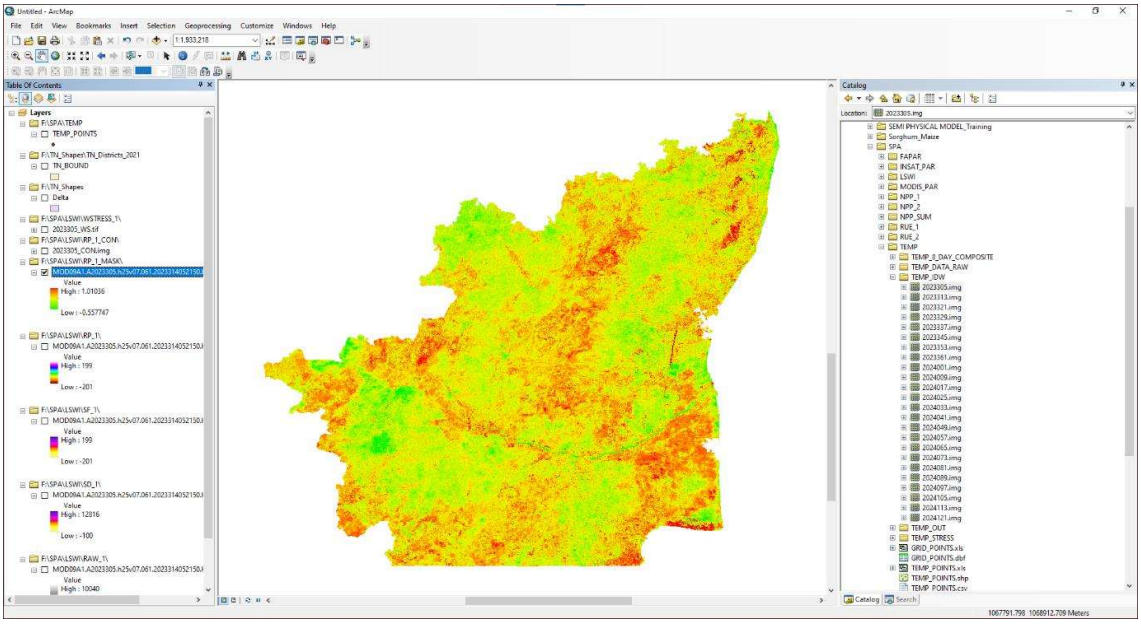


STEP IV: Calculation of LSWI from NIR & SWIR bands using *Raster Calculator* tool

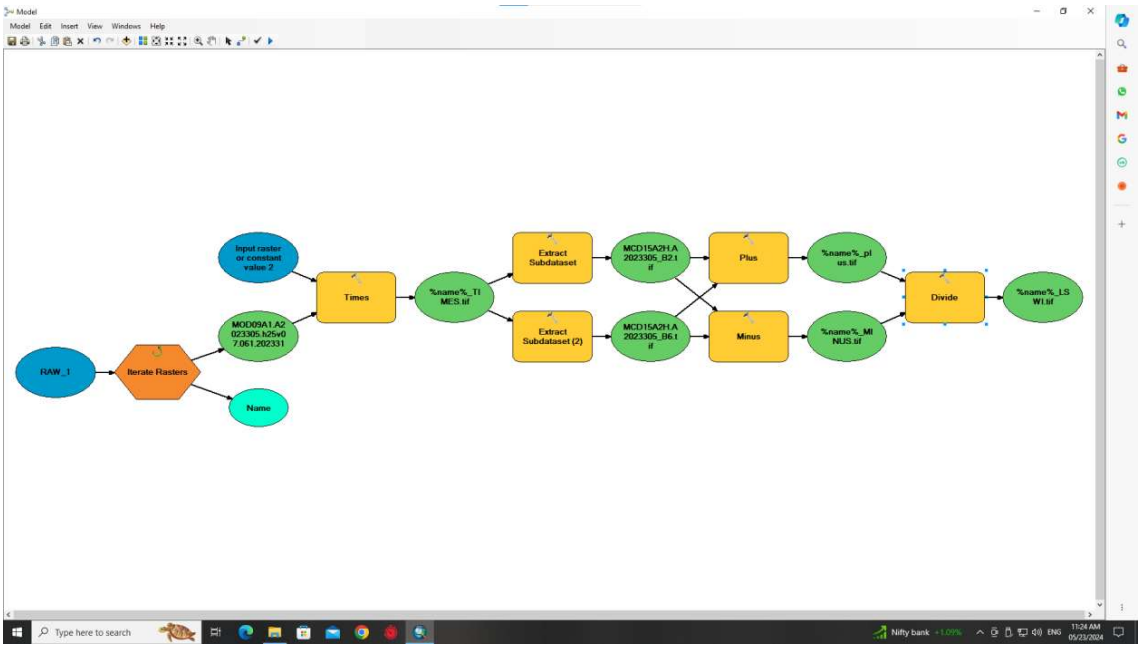
LSWI = (NIR-SWIR)/ (NIR+SWIR)



Raster Output of LSWI

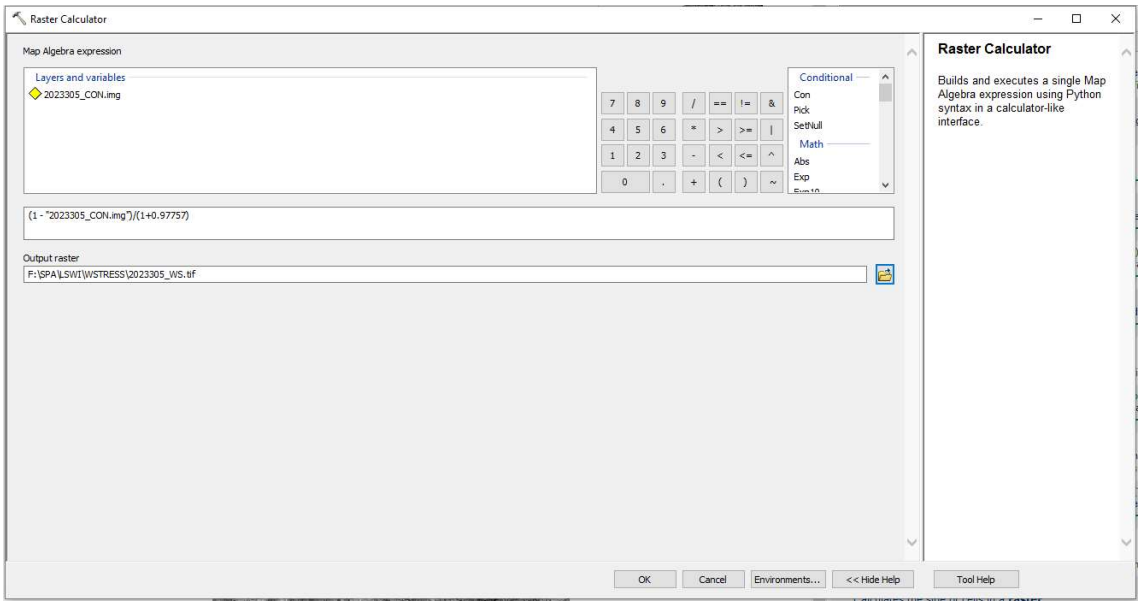


Generating Model for retrieving Land Surface Water Index from MOD09A1 using *Arc Map ModBuilder*

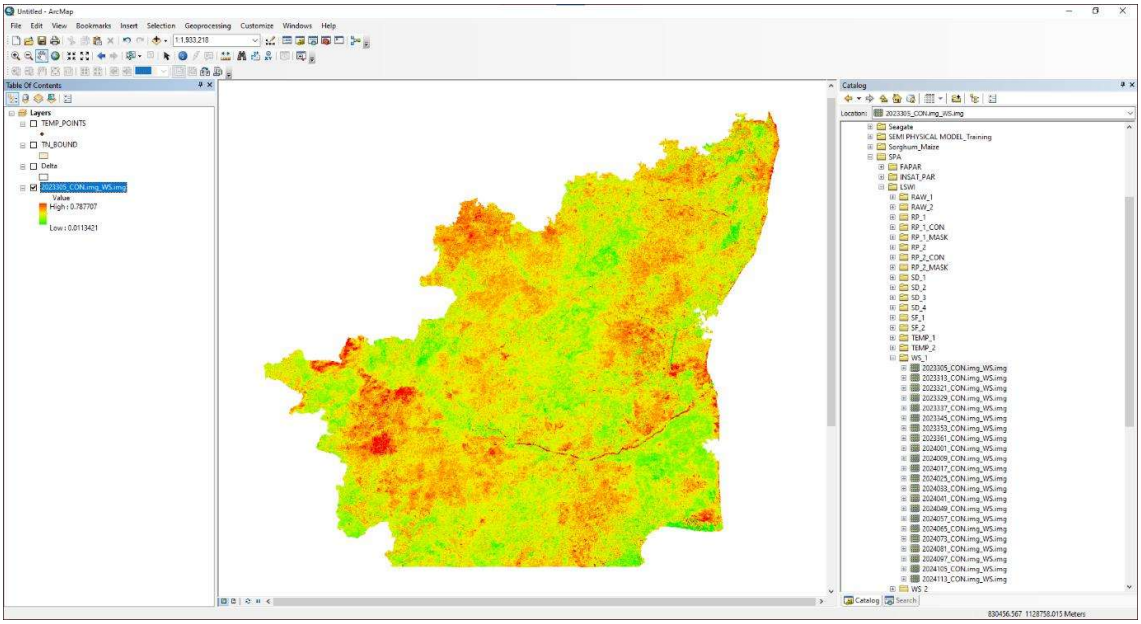


STEP V: Retrieving Water Stress from Land Surface Water Index using *Raster Calculator* tool

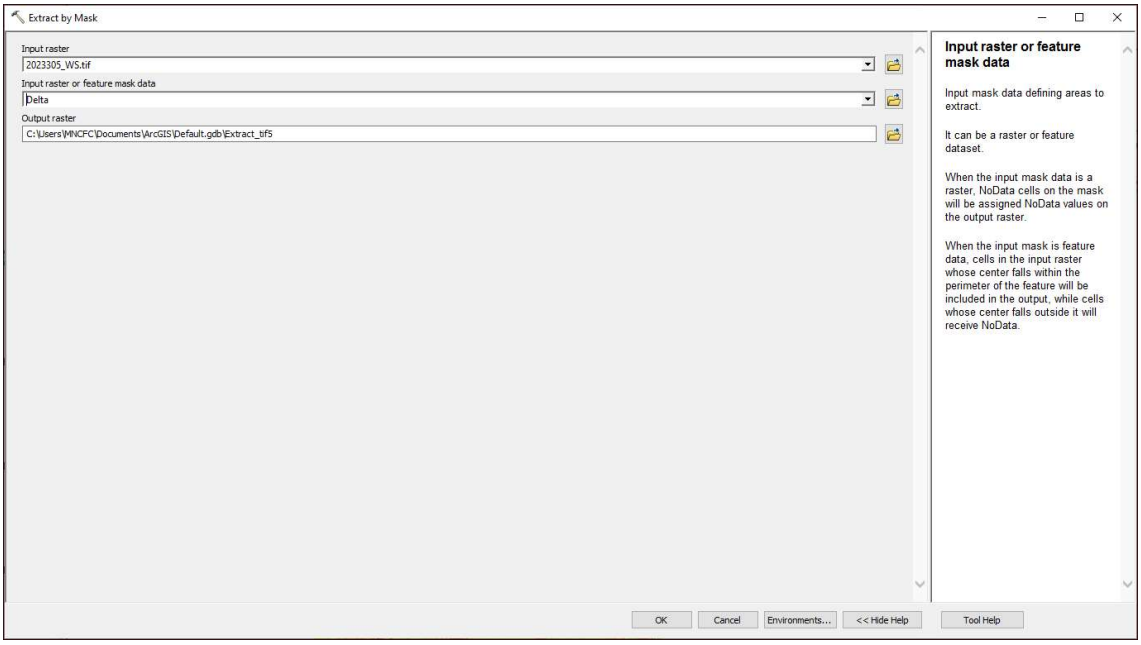
Water Stress = (1-LSWI)/(1+LSWI$_{max}$)



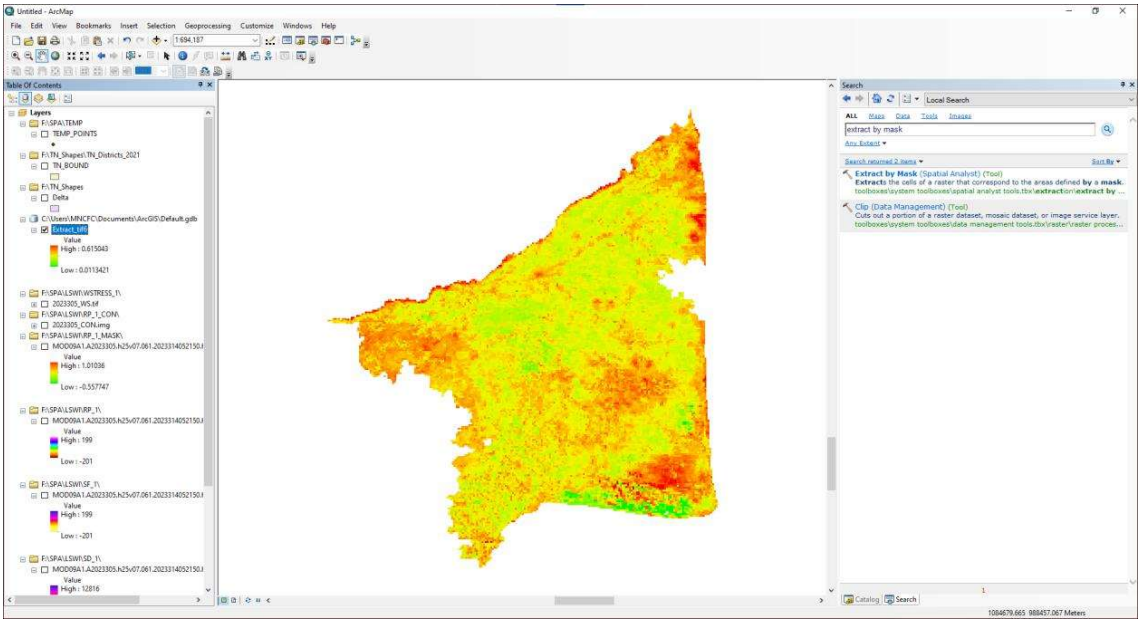
Raster Output of LSWI



STEP VI: Raster Extraction for study area using *Extract by Mask* tool

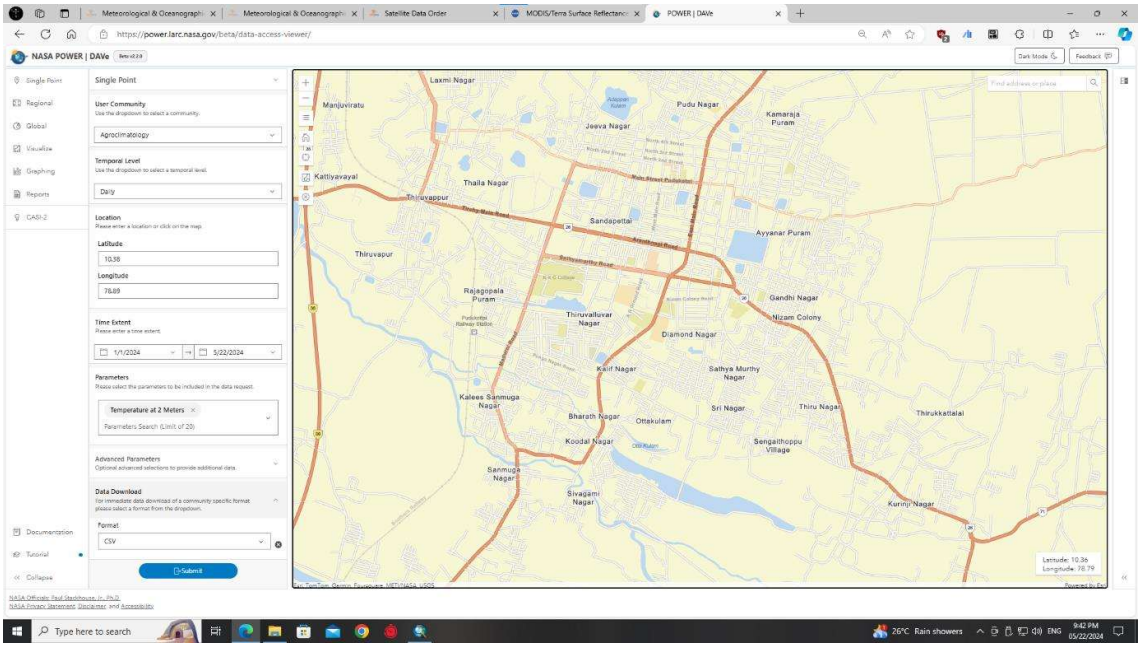


Raster output of Water Stress for the Study Area



D. Steps for generating Temperature Stress from Temperature data for study area

STEP I: Temperature Data acquisition from NASAPOWER portal



STEP II: CSV data sheet containing daily temperature data

